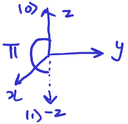
Properties of **Pauli operators**

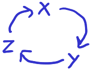
Bit flip $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$



$[I, Z] = 0$

Pauli vector operator: $\vec{\sigma} = (X, Y, Z) = (\sigma_x, \sigma_y, \sigma_z) = (\sigma_1, \sigma_2, \sigma_3)$

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$ in $B_z = \{|0\rangle, |1\rangle\}$



- Every Pauli operator is $\underbrace{\text{Hermitian}}$ (observable) $(X = X^\dagger)$ as well as unitary $(XX^\dagger = X^\dagger X = I)$: $X^2 = I$

$[X, Y] = XY - YX = 2iZ$

- Each Pauli operator has eigenvalues ± 1 and hence they are traceless $\mathrm{tr}(X) = 0$. $\mathrm{tr}(I) = 2$

$\{X, Y\} = XY + YX = 0$

- With the identity operator, they form the Pauli group under multiplication: $XY = \mathrm{i}\, Z$ $\{\pm 1, \pm i\}$

$YX = -iZ$

$\vec{a} \in \mathbb{R}^3$
$\vec{b} \in \mathbb{R}^3$

$$(\underbrace{\vec{a} \cdot \vec{\sigma}}_{\text{meas. spin in } \vec{a} \text{ dir}})(\vec{b} \cdot \vec{\sigma}) = (\vec{a} \cdot \vec{b})\, I + \mathrm{i}\,(\vec{a} \times \vec{b}) \cdot \vec{\sigma}$$

$\sum_k \sum_j a_j \sigma_j\, b_k \sigma_k$

$\sigma_j \sigma_k = i\, \sigma_l$ $\qquad \sigma_k \sigma_j = -i\, \sigma_l$

- With the identity operator, they form an orthogonal operator basis $\mathfrak{B} = \{I, X, Y, Z\}/\sqrt{2}$

Hilbert-Schmidt Inner product $[[X, Y]] := \mathrm{tr}(\underbrace{X^\dagger}_{X} Y) = 0\,, \quad [[X, X]] := \mathrm{tr}(\underbrace{X^\dagger X}_{I}) = 2$

$M_{2\times 2}$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\rho = \frac{1}{2} \sum_{k=0}^{3} \underbrace{[[\sigma_k, \rho]]}_{\langle \sigma_k \rangle}\, \sigma_k = \frac{I + \vec{r} \cdot \vec{\sigma}}{2}$$

$\langle v|u\rangle = v^\dagger u$

$|\psi\rangle = \sum \langle a_i|\psi\rangle\, |a_i\rangle$

$\vec{r} = (x, y, z) = (\langle X\rangle, \langle Y\rangle, \langle Z\rangle)$

$[[\sigma_1, \rho]] = \mathrm{tr}(\sigma_1 \rho) = \langle \sigma_1 \rangle$

$\langle I\rangle = \mathrm{tr}(\rho I) = \mathrm{tr}(\rho) = 1$

video

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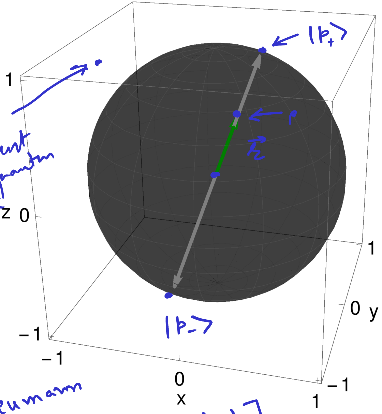
Single-qubit state ρ

Spectral decomposition of a

$\vec{r} = (x, y, z) \in \mathbb{R}^3$



does NOT represent any quantum state

$$\vec{r}\cdot\vec{\sigma} \equiv \begin{bmatrix} z & x-iy \\ x+iy & -z \end{bmatrix} \text{ in } B_z = \{|0\rangle, |1\rangle\}$$

$$\lambda^2 - \cancel{tr(\vec{r}\cdot\vec{\sigma})}\,\lambda + \underbrace{det(\vec{r}\cdot\vec{\sigma})} = 0$$

$tr(\vec{r}\cdot\vec{\sigma}) \to 0$

$$\lambda^2 - \underbrace{(x^2+y^2+z^2)}_{|\vec{r}|^2} = 0$$

$\lambda_\pm = \pm|\vec{r}|$

$$\rho^\dagger = \rho = \frac{I + \vec{r}\cdot\vec{\sigma}}{2}$$

$$= \underbrace{\left(\frac{1+|\vec{r}|}{2}\right)^{k}}_{p_+}\underbrace{\left(\frac{I+\hat{r}\cdot\vec{\sigma}}{2}\right)}_{|p_+\rangle\langle p_+|} + \underbrace{\left(\frac{1-|\vec{r}|}{2}\right)^{k}}_{p_-}\underbrace{\left(\frac{I-\hat{r}\cdot\vec{\sigma}}{2}\right)}_{|p_-\rangle\langle p_-|}$$

ρ^k

ρ is already normalized: $tr(\rho) = 1$

Positivity of $\rho \geq 0$ requires: $\underline{|\vec{r}| \leq 1}$

$\boxed{0 \le p_-} \le p_+$

$0 \le 1 - |\vec{r}|$

$|\vec{r}| \le 1$

Von Neumann

$S(\rho) = -[p_+ \ln p_+ + p_- \ln p_-]$

$= 0$ for a pure state $\rho^2 = \rho$

$\rho^2 = \rho$, $tr(\rho) = 1 \to tr(\rho^2) = 1$

video

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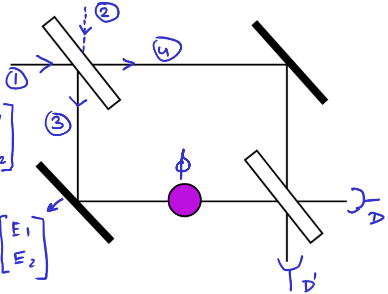
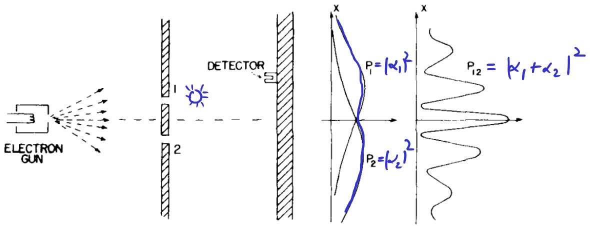


$$\begin{bmatrix} E_3 \\ E_4 \end{bmatrix} = \begin{bmatrix} r & t \\ t & -r \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}$$

$$\begin{bmatrix} E_4 \\ E_3 \end{bmatrix} = \begin{bmatrix} t & -r \\ r & t \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}$$

$t = r = \frac{1}{\sqrt{2}}$

$|t|^2 + |r|^2 = 1$

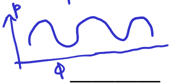


$$|\psi\rangle = \alpha_1|1\rangle + \alpha_2|2\rangle \qquad \{|1\rangle, |2\rangle\} \qquad \{|1\rangle \pm |2\rangle\}$$

Z X

$$e^{i\frac{\phi}{2}} \begin{pmatrix} \cos\frac{\phi}{2} \\ i\sin\frac{\phi}{2} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}}_{\text{Phase shifter}} \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}}_{\text{50-50 Beamsplitter}} \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\text{input}}$$

$p = |\cos\phi/2|^2$



$$= e^{i\frac{\phi}{2}} \begin{pmatrix} \cos\frac{\phi}{2} & i\sin\frac{\phi}{2} \\ i\sin\frac{\phi}{2} & \cos\frac{\phi}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H = H^{-1}$$

R. P. Feynman, *The Feynman Lectures on Physics*, Vol 3, Chapt 1.
M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Chapt 7.